

# **Python & SQL Data Analysis Project**

## **1. Project Name**

Target Database SQL & Python Analysis

## **2. Project Description**

This project analyses customer, order, payment, seller, product, and geolocation data to identify business trends and generate meaningful insights. It combines SQL-based data analysis with Python-based data manipulation and visualization to explore business performance and purchasing patterns.

## **3. Project Objectives**

- Analyse order trends and business performance.
- Examine order status and payment methods.
- Explore customer, seller, and product information.
- Identify trends in product categories and sales.
- Present analytical findings through visualizations.

## **4. Tools & Technologies**

- SQL
- MySQL Workbench
- Python
- Pandas
- Matplotlib
- Google Colab

## **5. Methodology**

The datasets were explored and prepared for analysis. SQL queries were used to investigate business metrics and answer analytical questions. Python and Pandas were used for data manipulation and further analysis, while Matplotlib was used to create visualizations and communicate the findings.

## 6. Key Analysis Areas

- Order status analysis
- Payment method distribution
- Monthly order trends
- Customer and seller analysis
- Product category analysis
- Top 10 product categories

## 7. Business Impact

The analysis demonstrates how SQL and Python can be used to explore business data, identify trends, and support data-driven decision-making related to orders, payments, customers, sellers, and product categories.

## 8. GitHub Repository

[https://github.com/Mahanand4/Target\\_Database\\_SQL\\_Python\\_Analysis](https://github.com/Mahanand4/Target_Database_SQL_Python_Analysis)